The Travelling Salesman Problem

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Abstract

In permutation-based combinatorial optimisation problems such as the Travelling Salesman Problem (TSP), candidate solutions are represented as permutations of cities. Tour length depends only on the adjacency of cities (the undirected edges connecting them), rather than their absolute positions in the chromosome. This study investigates the following question: *Do crossover operators that preserve parental edges perform better than those that preserve absolute positions, and is edge heritability the main reason?* Using the TSPLIB benchmark instance `ulysses22` ($n = 22$ cities, geographical distance metric, global optimum $L^* = 7013$), we compare three classic crossover operators: Partially Mapped Crossover (PMX, preserving absolute positions through mappings), Order Crossover (OX, preserving relative order and contiguous sub-paths), and Edge Recombination (ER, prioritising parental edge adjacencies). We also include a baseline control without crossover (NO-CX), which uses selection, inversion mutation, and elitism. Using a controlled matched-seed design ($R = 30$ independent runs per configuration, $G = 1{,}000$ generations, population $N = 200$), the overall Friedman test found significant performance differences ($\chi^2 = 18.3024$, $p = 0.000381$). Post-hoc Wilcoxon signed-rank tests with Holm step-down correction showed that both OX ($p_{\text{adj}} = 0.0341$) and ER ($p_{\text{adj}} = 0.0034$) significantly outperformed PMX in final tour length. ER also significantly outperformed the NO-CX baseline ($p_{\text{adj}} = 0.0345$). However, no statistically significant difference was found between OX and ER ($p_{\text{adj}} = 0.7932$), or between OX and NO-CX after Holm correction ($p_{\text{adj}} = 0.1199$). These non-significant results do not prove that the configurations are equivalent; they indicate that there is insufficient evidence to establish a difference. Tracking dynamic edge heritability (h_e) and population edge diversity ($D_e(t)$) showed that ER achieved the highest edge preservation ($h_e = 0.9991$), but also experienced a rapid loss of population diversity ($D_e \approx 4.09$ by generation 100). In contrast, OX maintained substantially higher population edge diversity ($D_e \approx 7.50$ at generation 100), balancing exploration and exploitation. This allowed OX to achieve similar final results to ER, with optimum success rates of 60.0% for OX and 56.7% for ER. Overall, the results show that preserving parental edges is more effective than preserving absolute positions when compared with PMX. However, edge heritability alone does not explain optimisation success. Edge preservation must be balanced with population diversity to reduce the risk of premature stagnation.

1 Introduction, Research Question, and Hypotheses

The Travelling Salesman Problem (TSP) is a well-known NP-hard problem in combinatorial optimisation [12, 13]. Given a set of n cities and the distances between every pair, the objective is to find a closed Hamiltonian tour that visits each city exactly once and returns to the starting city while minimising the total travel distance.

When solving the TSP using Genetic Algorithms (GAs), candidate tours are naturally represented as permutations of city identifiers $[0, 1, \ldots, n-1]$. Standard binary crossover operators (such as single-point, two-point, or uniform crossover) cannot be directly applied because they can produce invalid offspring with duplicated and missing cities [6, 11]. Therefore, specialised permutation crossover operators have been developed. These operators differ in the type of information they preserve from the parent tours:

1. **Positional Operators:** Operators such as Partially Mapped Crossover (PMX) [7] focus on preserving the positions of cities in the parent chromosomes. They use mappings between parents to resolve duplicate assignments.

2. **Order-Based Operators:** Operators such as Order Crossover (OX) [2] copy a contiguous sub-path from one parent and fill the remaining positions using the relative order of unvisited cities from the second parent.

3. **Edge-Based Operators:** Operators such as Edge Recombination (ER) [14] focus on preserving direct adjacencies (undirected edges) found in the parents. They build an edge map and select adjacent unvisited cities using a least-degree heuristic to reduce the introduction of new edges.

Despite these different designs, the cost of a TSP tour depends only on its edges (the connections between adjacent cities, including the connection between the last and first cities). It remains unchanged under cyclic shifts and tour reversals. Therefore, absolute positions in the chromosome have no meaning for a circular tour. This leads to the main research question of this study.

1.1 Research Question

Do crossover operators that pass on the edges of the parents work better than those that pass on positions, and is the heritability of edges the reason?

1.2 Hypotheses and Refutation Criteria

- **Hypothesis 1 (Edge Heritability):** Edge Recombination (ER) will achieve the highest dynamic edge heritability (h_e), followed by Order Crossover (OX) and Partially Mapped Crossover (PMX). Because ER explicitly queries parent edge lists during offspring construction, it is hypothesized to preserve significantly more parental edges than operators that treat tours as linear strings.

  Refutation criterion: Hypothesis 1 would be refuted if ER fails to attain higher measured dynamic edge heritability than OX and PMX, or if its edge heritability is not consistently superior across generations and independent runs.

- **Hypothesis 2 (Search Performance):** Operators that preserve parental edge information (OX and ER) will achieve shorter final tour lengths, lower run-to-run variability, and higher global-optimum success rates than PMX. This is expected because preserving valid low-cost edges aligns directly with the TSP objective function, whereas preserving absolute string positions disrupts critical adjacencies.

  Refutation criterion: Hypothesis 2 would be refuted if OX and ER do not attain statistically significantly shorter final tour lengths or higher global-optimum success rates than PMX under identical experimental conditions.

- **Mechanistic Link:** If higher edge heritability directly drives optimisation success, we expect a monotonic correspondence between heritability and final solution quality. If, however, high edge heritability does not correspond to superior search performance, or if operators with lower heritability perform comparably or better, this would indicate that edge heritability alone is insufficient to explain crossover effectiveness and that secondary dynamics (such as population diversity maintenance) govern GA convergence.

1.3 Operators and Control Configuration

Table 1 summarizes the operators compared in this study. We included an explicit baseline control. In NO-CX, crossover is completely disabled ($p_c = 0.0$); the algorithm relies exclusively on tournament selection, Simple Inversion Mutation ($p_m = 0.2$), and elitism ($E = 1$). NO-CX serves as a baseline to isolate and measure the contribution of crossover recombination to mutation-driven search.

Table 1: Permutation crossover operators and baseline control evaluated on `ulysses22`.

Config	Preserved Property	Reference	Operational Description
PMX	Position & Mapping	Goldberg & Lingle (1985) [7]	Copies an enclosed segment from Parent 1; determines matching alleles in Parent 2 and uses bidirectional mapping to fill remaining positions without duplication.
OX	Relative Order	Davis (1985) [2]	Copies a contiguous segment from Parent 1; preserves the relative linear sequence of the remaining cities as they appear in Parent 2, starting after the cut.
ER	Undirected Edges	Whitley et al. (1989) [14]	Constructs an edge map listing all neighbours of each city across both parents; constructs offspring by iteratively selecting unvisited neighbours with minimal remaining degree.
NO-CX	None (Control)	Baseline Control	Crossover is disabled ($p_c = 0.0$). Generates offspring purely via tournament selection, inversion mutation ($p_m = 0.2$), and elitism. Isolates the net contribution of crossover.

2 Experimental Design and Methodology

2.1 Benchmark Problem: TSPLIB `ulysses22`

We evaluate the algorithms on the standard TSPLIB benchmark instance `ulysses22.tsp` [13]. This dataset consists of $n = 22$ geographical locations representing the Mediterranean voyage of Odysseus in Homer's *Odyssey*, compiled and optimized by Grötschel and Padberg [8].

Distances are computed using the exact TSPLIB `GEO` distance formula: coordinates given in degrees and minutes are converted to radians, and great-circle distances on an idealized sphere ($R = 6378.388$ km) are rounded to

integers. The verified global optimum tour distance for `ulysses22` under TSPLIB `GEO` rules is:

$$L^* = 7013 \tag{1}$$

2.2 Controlled Factorial Design (One-Knob Principle)

We follow the single-factor manipulation principle: the only algorithmic factor deliberately varied across configurations is the **recombination strategy** (PMX, OX, ER, and the NO-CX baseline control). All other algorithmic hyperparameters, representations, data structures, selection mechanisms, mutation operators are kept constant across configurations.

Table 2: Fixed genetic algorithm parameters and justifications.

Parameter	Value	Scientific Justification
Instance	`ulysses22` ($n = 22$)	Core problem instance; known global optimum $L^* = 7013$.
Representation	Permutation of length $n = 22$	Natural, non-redundant representation for TSP tours.
Population size (N)	200 individuals	Sufficiently large to prevent immediate genetic drift on $n = 22$.
Generations (G)	1,000 generations	Budget of 200,000 evaluations per run, ensuring convergence.
Selection scheme	Tournament selection ($k = 3$)	Balances selection pressure with diversity preservation without scale sensitivity.
Crossover probability (p_c)	0.90 (0.0 for NO-CX)	Standard high crossover rate recommended in literature [6].
Mutation operator	Simple Inversion Mutation (SIM)	Reverses a random sub-tour; natural mutation for TSP (disrupts only 2 edges).
Mutation probability (p_m)	0.20 per individual	Moderate mutation rate maintaining continuous exploration throughout search.
Elitism	1 individual (best-so-far)	Strict preservation of the hall-of-fame solution across generations.
Evaluation budget	200,000 tour evaluations	Identical computational effort across all runs and configurations.
Random seed blocking	Seeds 2000–2029 ($R = 30$)	Identical initial populations per seed across all 4 configurations (paired design).

2.3 Quantitative Mechanistic Metrics

To uncover the internal evolutionary mechanics beyond final fitness scores, two fine-grained metrics are tracked dynamically every 10 generations across all runs:

2.3.1 Dynamic Edge Heritability (h_e)

Edge heritability (h_e) measures the fraction of offspring edges that were present in at least one of the two parents. Heritability is recorded immediately after crossover and *before mutation*, allowing us to measure the effect of the crossover operator without including changes caused by mutation.

We track the rolling mean of the last 50 crossover events at each 10-generation checkpoint. For the NO-CX baseline, crossover is not used, so edge heritability is not applicable (`N/A`).

2.3.2 Population Edge Diversity ($D_e(t)$)

Population edge diversity ($D_e(t)$) measures how many different undirected edges exist across all individuals in the population at generation t, relative to the tour length n.

A value of 1.0 is the theoretical minimum. This occurs when all individuals have exactly the same cyclic edges.

For $n = 22$ cities, the theoretical maximum is 10.5. This is because the complete graph K_{22} contains 231 unique undirected edges, and dividing this by the 22 edges in each tour gives 10.5.

2.4 Pilot Study and Sample Size Justification

Before executing the full experiment, we conducted a preliminary pilot study with $R = 10$ runs per operator using separate seeds (1000–1009) to observe the variation between runs. The results (Table 3) showed differences in tour length across operators. PMX had the highest variation (standard deviation $\sigma = 63.44$) and found the global optimum in only 30% of runs. In comparison, OX ($\sigma = 40.41$) and ER ($\sigma = 38.30$) achieved success rates of 70% and 80%, respectively.

Sample Size Selection: Based on the variance observed in the pilot experiments, we selected $R = 30$ runs for the main study. The same seeds (2000–2029) were used for all four configurations, ensuring that each algorithm started with the same initial random populations. This allowed us to compare the algorithms under the same starting conditions and reduced the effect of differences between initial populations. It also improved the sensitivity of the paired non-parametric tests.

Table 3: Pilot study summary ($R = 10$ independent runs, seeds 1000–1009).

Operator	Runs	Median	IQR	Mean	SD	Min	Max	Opt Rate (%)	Censored (%)
PMX	10	7091.5	86.75	7089.2	63.44	7013	7190	30.0%	70.0%
OX	10	7013.0	59.25	7038.0	40.41	7013	7105	70.0%	30.0%
ER	10	7013.0	0.00	7031.0	38.30	7013	7114	80.0%	20.0%

3 Experimental Results and Comparative Analysis

The main experiment comprises 120 complete GA runs (30 matched runs for each of PMX, OX, ER, and NO-CX).

Table 4: Main experiment performance summary on `ulysses22` ($R = 30$ matched runs, $G = 1{,}000$ generations). Censored runs are those failing to reach $L^* = 7013$ within the 1,000-generation budget.

Operator	Runs	Median	IQR	Mean	SD	Min	Max	Opt (%)	Med Gen	Cens (%)	Final D_e
PMX	30	7107.0	107.00	7113.40	72.89	7013	7241	23.33%	1000.0	76.67%	4.12
OX	30	7013.0	95.50	7052.77	51.77	7013	7163	60.00%	221.5	40.00%	4.36
ER	30	7013.0	79.00	7051.43	46.48	7013	7150	56.67%	52.5	43.33%	3.97
NO-CX	30	7089.5	132.25	7090.27	75.50	7013	7241	40.00%	1000.0	60.00%	4.03

3.1 Performance Observations

- **Global Optimum Success Rate:** OX achieved the highest success rate at 60.00% (18 out of 30 runs reaching $L^* = 7013$), closely followed by ER at 56.67% (17 out of 30 runs). In contrast, PMX reached the optimum in only 23.33% of runs (7 out of 30), suffering a censoring rate of 76.67%. The NO-CX baseline reached the global optimum in 40.00% of runs (12 out of 30).
- **Central Tendency and Dispersion:** Both OX and ER achieved a median final tour length of 7013.0 (the global optimum), with tightly clustered means of 7052.77 and 7051.43, respectively. ER demonstrated the lowest overall standard deviation ($\sigma = 46.48$) and lowest interquartile range (IQR = 79.00). Conversely, PMX exhibited the worst median (7107.0) and highest mean (7113.40), while NO-CX yielded a median of 7089.5 and a high dispersion ($\sigma = 75.50$, IQR = 132.25).
- **Convergence Speed:** Among the runs that discovered the global optimum, ER converged remarkably early, with a median discovery generation of 52.5, compared to 221.5 for OX. For PMX and NO-CX, the median generation of first optimum is recorded as 1000.0 because over half of their runs were censored (i.e., did not attain the optimum before the budget expired).

Figure 1 illustrates the final tour length distributions and convergence dynamics across the 120 runs.

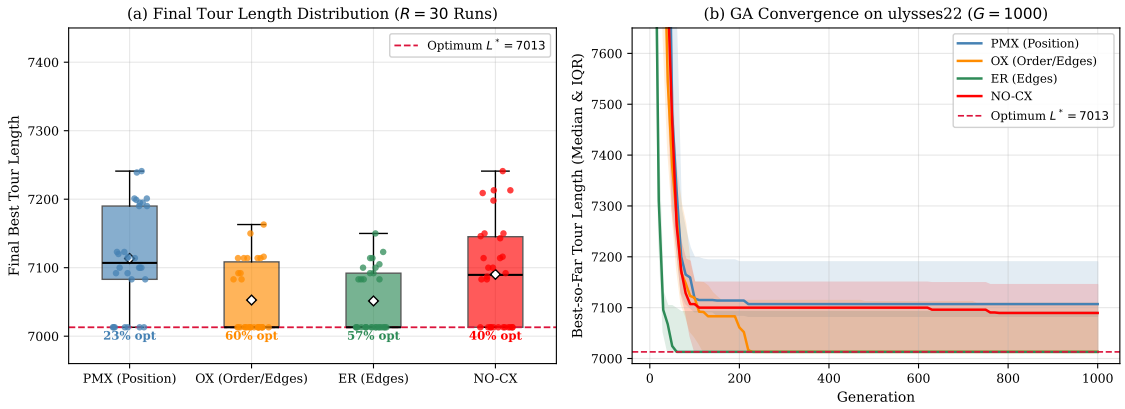


Figure 1: (a) Boxplots of final tour length distributions with individual runs jittered, diamond markers representing configuration means, and success rate labels. (b) Best-so-far tour length over 1,000 generations showing median trajectories with shaded interquartile ranges (IQR, 25th–75th percentiles).

4 Rigorous Statistical Analysis

All four configurations used the same 30 random seeds (2000–2029), allowing us to compare their results in pairs. Since the results do not meet the assumptions of parametric normality, we used non-parametric paired tests. [1].

4.1 Omnibus Test: Friedman Non-Parametric Test

We first apply the Friedman test across all four configurations across the 30 matched blocks to test the null hypothesis that all four configurations share identical performance distributions:

$$\chi^2 = 18.3024, \quad \text{df} = 3, \quad p = 0.000381 \tag{2}$$

Because $p < 0.001$, the null hypothesis of equal performance is decisively rejected at $\alpha = 0.05$. This justifies conducting post-hoc pairwise comparisons to determine which specific configurations differ.

4.2 Pairwise Wilcoxon Signed-Rank Tests and Effect Sizes

We evaluate all 6 pairwise comparisons using the two-sided Wilcoxon signed-rank test [15]. In accordance with standard statistical methodology:

1. **Exclusion of Ties:** Matched pairs with zero difference ($d_i = 0$) are excluded from the ranking calculation following the Wilcoxon convention. The number of ties is explicitly reported.
2. **Multiplicity Correction:** To strictly control the family-wise error rate at $\alpha = 0.05$, Holm's step-down sequentially rejective procedure [10] is applied across all six comparisons.
3. **Hodges–Lehmann Median Paired Difference:** We report the Hodges–Lehmann (HL) estimator [9], computed as the median of all pairwise Walsh averages $(d_i + d_j)/2$.
4. **Bootstrap Confidence Intervals:** For each HL difference, we report a 95% paired percentile bootstrap confidence interval constructed from $B = 5{,}000$ resamples with replacement.
5. **Matched-Pairs Rank-Biserial Correlation ($r_{\mathbf{rb}}$):** Positive values indicate that Configuration A produces longer (worse) tours than Configuration B.
6. **Win / Loss / Tie Counts:** We report the raw counts and win fractions among non-tied pairs.

Table 5 presents the comprehensive statistical outcomes.

Table 5: Pairwise statistical comparisons across 30 matched seeds ($B = 5{,}000$ bootstrap resamples). Significant outcomes after Holm step-down correction ($\alpha = 0.05$) are marked in bold.

Comparison	HL Diff	95% Bootstrap CI	$r_{\mathbf{rb}}$	Wins/Loss/Tie	Win Frac	W	$p_{\mathbf{raw}}$	$p_{\mathbf{adj}}$	Signif?
PMX vs OX	57.5	[22.5, 101.0]	0.5952	7 / 20 / 3	0.2593	76.5	0.0068	**0.0341**	**Yes**
PMX vs ER	59.0	[35.0, 93.0]	0.8033	3 / 21 / 6	0.1250	29.5	0.0006	**0.0034**	**Yes**
PMX vs NO-CX	35.0	[−11.5, 64.0]	0.2821	9 / 17 / 4	0.3462	126.0	0.2086	0.4172	No
OX vs ER	0.0	[−35.0, 29.0]	0.0667	9 / 11 / 10	0.4500	98.0	0.7932	0.7932	No
OX vs NO-CX	−42.0	[−68.5, 0.0]	−0.5108	15 / 6 / 9	0.7143	56.5	0.0400	0.1199	No
ER vs NO-CX	−35.0	[−65.0, −1.0]	−0.6537	16 / 5 / 9	0.7619	40.0	0.0086	**0.0345**	**Yes**

4.3 Interpretation of Statistical Findings

The statistical results show several important findings:

1. **OX and ER Perform Significantly Better than PMX:** Both OX ($p_{\text{adj}} = 0.0341$) and ER ($p_{\text{adj}} = 0.0034$) achieved significantly shorter tour lengths than PMX. Their 95% bootstrap confidence intervals for the HL median difference were positive. Both also showed large effect sizes ($r_{\text{rb}} = 0.5952$ and 0.8033, respectively). In direct comparisons without tied results, OX won 20 out of 27 matches against PMX (74.1%), while ER won 21 out of 24 matches (87.5%).
2. **ER Performs Significantly Better than NO-CX:** ER achieved significantly better results than the GA without crossover ($p_{\text{adj}} = 0.0345$, $r_{\text{rb}} = -0.6537$). It also won 16 out of 21 non-tied comparisons. The 95% bootstrap confidence interval for ER minus NO-CX was negative ($[-65.0, -1.0]$). This confirms that edge recombination provides a genuine evolutionary advantage over mutation-only search

3. **PMX Does Not Perform Better than NO-CX:** There was no statistically significant difference between PMX and NO-CX ($p_{\text{adj}} = 0.4172$). The bootstrap confidence interval also crossed zero ($[-11.5, 64.0]$). PMX actually suffered more losses than wins against NO-CX (9 wins vs. 17 losses), indicating that position-based crossover can be actively counter-productive on the TSP.

4. **OX vs. ER and OX vs. NO-CX:** The comparison between OX and ER produced a Hodges–Lehmann difference of 0.0 and an adjusted p-value of 0.7932 ($r_{\text{rb}} = 0.0667$). The comparison between OX and NO-CX had an unadjusted $p = 0.0400$, but after Holm correction it was no longer statistically significant ($p_{\text{adj}} = 0.1199$). Its 95% bootstrap confidence interval was $[-68.5, 0.0]$. **Crucially, these non-significant outcomes do not prove equivalence.** In statistical hypothesis testing, failure to reject the null hypothesis merely indicates that the observed sample size ($R = 30$) and inherent stochastic variance provide insufficient evidence to detect a difference at the chosen error rate.

5 Mechanistic Analysis: Heritability vs. Diversity

Why did OX and ER perform better than PMX? And why did OX achieve similar results to ER even though ER focuses directly on preserving edges? To answer this, we examine the dynamic evolutionary curves of dynamic edge heritability (h_e) and population edge diversity ($D_e(t)$) depicted in Figure 2.

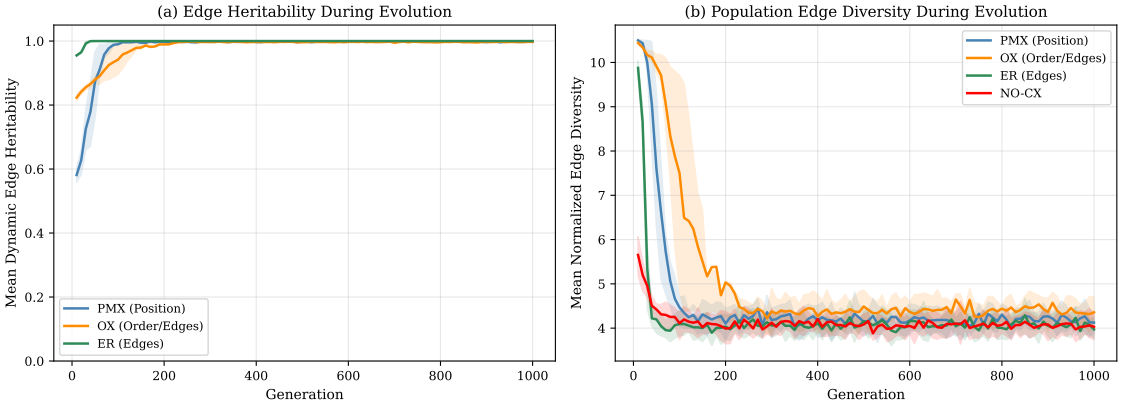


Figure 2: (a) Mean dynamic edge heritability (h_e) recorded immediately after crossover and prior to mutation (solid line: mean; shaded band: interquartile range). NO-CX is omitted because crossover is disabled. (b) Mean normalized population edge diversity ($D_e(t)$) across all four configurations.

5.1 Dynamic Edge Heritability Dynamics

Table 6 summarizes the heritability and diversity values across key evolutionary milestones.

As predicted by Hypothesis 1, Edge Recombination achieves the highest mean dynamic edge heritability across evolution:

$$\bar{h}_e(\text{ER}) = 0.999122 \approx 99.91\% \tag{3}$$

ER uses the combined edges from both parents and selects edges using the least-degree heuristic. As a result, it rarely introduces new edges unless it reaches a dead end and must make a random choice.

Order Crossover achieves a mean dynamic edge heritability of:

$$\bar{h}_e(\text{OX}) = 0.983985 \approx 98.40\% \tag{4}$$

Partially Mapped Crossover achieves the lowest mean dynamic edge heritability:

$$\bar{h}_e(\text{PMX}) = 0.981972 \approx 98.20\% \tag{5}$$

Although all three operators exhibit heritability above 98% in later generations (as the population becomes increasingly homogeneous), their early-stage behavior differs markedly: in early generations, PMX frequently breaks parental adjacencies because it maps individual positions rather than contiguous sub-paths or edges.

5.2 Population Edge Diversity Dynamics

The diversity results (Figure 2b and Table 6) help explain why ER's high edge heritability did not lead to a higher success rate than OX:

Table 6: Evolutionary trajectory of mean population edge diversity ($D_e(t)$) and overall mean edge heritability (h_e) across $R = 30$ runs on `ulysses22`.

	Overall Mean	Mean Population Edge Diversity (D_e) at Generation t							Overall Mean
Config	**Heritability (h_e)**	**Gen 10**	**Gen 100**	**Gen 200**	**Gen 400**	**Gen 600**	**Gen 800**	**Gen 1000**	**Diversity (D_e)**
PMX	0.9820	10.5000	4.4682	4.1030	4.1788	4.1742	4.3076	4.1242	4.5222
OX	0.9840	10.4409	7.5030	5.0333	4.2697	4.3667	4.4394	4.3576	**5.0207**
ER	**0.9991**	9.8773	4.0924	4.0364	4.1091	4.1636	4.0030	3.9697	4.1857
NO-CX	*N/A*	5.6530	4.1121	3.9894	4.2076	4.0500	4.0712	4.0303	4.1272

- **OX Maintains Higher Diversity:** OX keeps more edge diversity throughout the search, with an overall mean of 5.0207. At generation 100, its diversity is 7.5030, compared to 4.4682 for PMX, 4.1121 for NO-CX, and 4.0924 for ER. This allows OX to explore different edge combinations for longer and continue finding better sub-tours after 200 generations.
- **ER Loses Diversity Quickly:** ER combines parental edges very effectively and achieves almost 100% edge heritability. However, its edge diversity drops quickly. By generation 100, it reaches 4.0924, which is close to the NO-CX baseline (4.1121). This helps explain ER's different results across runs. When the initial population contains the optimal edges, ER can find the optimum quickly (median discovery generation 52.5). However, when some optimal edges are missing, its low diversity makes it harder to discover new edges, which can lead to getting stuck in a suboptimal solution.
- **PMX Has Low Diversity and Edge Preservation:** PMX also loses diversity quickly, reaching 4.4682 at generation 100. However, unlike ER, it does not preserve parental edges as effectively. This results in the weakest final search performance.
- **NO-CX Behavior:** Without crossover, NO-CX loses diversity quickly. By generation 10, its diversity drops to 5.6530, compared to 9.88–10.50 for the crossover configurations. It depends entirely on inversion mutation to introduce new edges. This gives it a moderate success rate of 40%, but it cannot match ER's speed in finding the optimum.

6 Geographical Tour Visualisation

Figure 3 shows the global optimum tour and an example of a suboptimal tour on `ulysses22`. This helps illustrate how changing connections between cities can affect the total tour length.

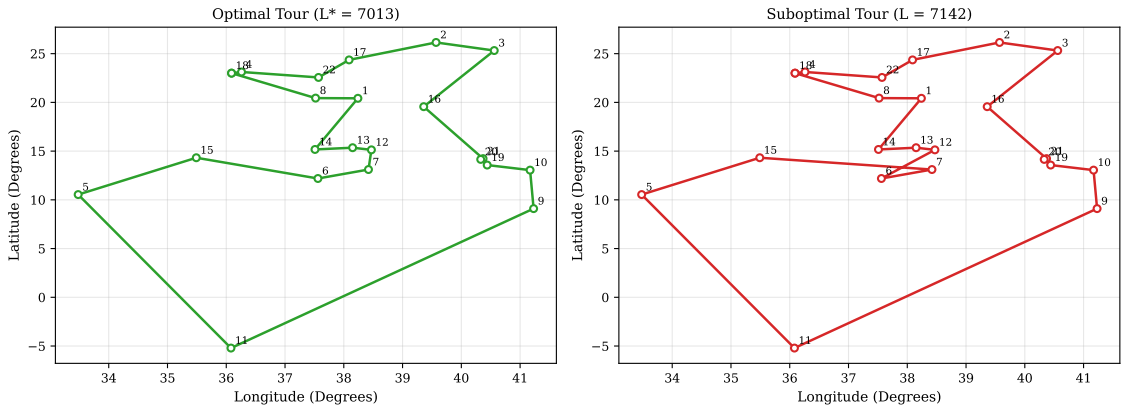


Figure 3: (a) Known global optimum tour ($L^* = 7013$). (b) Manually created suboptimal tour ($L = 7142$)

The known global optimum tour sequence (0-indexed city identifiers) is:

$$T^* = [17, 3, 21, 16, 1, 2, 15, 20, 19, 18, 9, 8, 10, 4, 14, 5, 6, 11, 12, 13, 0, 7] \tag{6}$$

Its total tour length is exactly $L(T^*) = 7013$.

To show how positional crossover, such as PMX, can affect connections in a cyclic TSP tour, consider the following manually created example:

$$T_{\text{sub}} = [17, 3, 21, 16, 1, 2, 15, 20, 19, 18, 9, 8, 10, 4, 14, \mathbf{6}, \mathbf{5}, 11, 12, 13, 0, 7] \tag{7}$$

In this example, the positions of cities 5 and 6 are swapped. Although 20 of the 22 cities remain in the same positions, this change breaks three cyclic edges ($\{14, 5\}$, $\{5, 6\}$, $\{6, 11\}$). These connections are replaced by longer edges, increasing the total tour length by 129 units, from 7013 to 7142.

7 Threats to Validity and Study Limitations

1. **Instance Size and Topology:** Our experiments were carried out only on `ulysses22` ($n = 22$), as required for the main study. Its relatively small size means that the search space is explored using a population of $N = 200$ over 1,000 generations. Larger TSPLIB instances, such as `kroA100` and `pr2392`, may show different results. Therefore, our findings should not automatically be applied to larger instances without further testing [11, 14].

2. **Fixed Parameters:** We kept all parameters fixed ($N = 200$, $p_c = 0.9$, $p_m = 0.2$, tournament $k = 3$) to focus on the effect of the crossover operators. However, different parameter settings could produce different results. For example, a higher mutation rate might help reduce ER's loss of diversity, while a smaller tournament size could reduce selection pressure.

3. **No Local Search:** Our study used a standard Genetic Algorithm without local search methods such as 2-opt, 3-opt, or Lin–Kernighan. These methods are commonly used in more advanced TSP approaches [11]. Adding local search could change the results and reduce the differences between crossover operators.

4. **Correlation and Causality:** We observed relationships between edge heritability, population edge diversity, convergence speed, and search stagnation. However, these observations show correlations and do not prove that one directly causes the other.

8 Reproducibility Statement

All experimental results, statistical tests, tables, and figures can be reproduced using the provided code and data.

- **Environment and Dependencies:** The project was implemented in Python using `numpy`, `scipy` , `pandas`, `matplotlib`, `deap`, and `tsplib95`.

- **Random Seeds:** Pilot experiments used seeds 1000–1009 ($R = 10$), while the main experiments used seeds 2000–2029 ($R = 30$). Python's `random` and NumPy's `np.random` seeds were synchronised before each run.

- **Code and Results:** The code is available in the Jupyter notebook `tsp_study.ipynb`. Raw experimental results and statistical summaries are stored as CSV files in the `results` directory. Instructions for reproducing the experiments are provided in `README.txt`.

Generative AI Disclosure

ChatGPT and Gemini was used as a supplementary support tool during the project to clarify theoretical concepts, assist with troubleshooting technical issues, and improve the clarity of the written report. All suggestions were reviewed by us.

9 Conclusion

Our results answer the two research questions:

1. **Do edge-preserving operators perform better than position-preserving operators?** Yes. Both Order Crossover (OX) and Edge Recombination (ER) performed significantly better than Partially Mapped Crossover (PMX) on `ulysses22` ($p_{\text{adj}} = 0.0341$ and 0.0034, respectively). PMX did not perform better than the NO-CX baseline.

2. **Is preserving edges enough to achieve better results?** No. ER preserved almost all parental edges ($h_e = 0.9991$) and converged faster in the early stages. However, it also caused a large reduction in population edge diversity. OX maintained higher edge diversity ($D_e = 5.0207$ compared to 4.1857 for ER), allowing more exploration. Both operators achieved similar global optimum success rates (60% for OX and 56.7% for ER).

Overall, our results show that preserving edges is more effective than preserving absolute positions when solving the TSP. However, preserving edges alone is not enough. Maintaining diversity in the population is also important for finding good solutions.

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